

AI-Powered University Timetable Management System: Design and Implementation of an AI-Assisted Centralized Management Platform

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Abstract

University timetable management is a complex administrative task that requires the coordination of teachers, courses, semesters, sections, and time slots across multiple academic departments. In many university environments, timetable preparation is performed independently by departments, which can make coordination difficult when faculty members teach courses across different departments. This decentralized approach may result in scheduling conflicts, particularly when the same teacher is assigned to multiple classes during the same time period.

This paper presents an AI-assisted university timetable management system designed to provide centralized management and coordination of departmental timetables. The proposed system follows a role-based architecture in which Super Administrators, Deans, and Chairmen perform different administrative and academic management tasks, while timetable approval is coordinated through a structured workflow. The system provides centralized management of teachers, courses, semesters, sections, and timetable slots, while rule-based validation is used during timetable creation to identify teacher and batch/section scheduling conflicts.

To improve interaction with timetable information, the system incorporates an AI-assisted query mechanism using the Google Gemini API. Natural-language queries are interpreted to extract relevant intent and entities, such as department, academic year, semester, section, and teacher name. The extracted information is then processed by the backend to execute deterministic queries against a MongoDB database and return the corresponding timetable information. This separates natural-language interpretation from database operations and maintains application-level control over the retrieved data.

The resulting architecture combines centralized timetable administration, role-based access control, rule-based conflict validation, approval management, and AI-assisted information retrieval within a single web-based platform. The system demonstrates how AI can be integrated as an assistance layer within a conventional timetable management architecture

without relying on AI to autonomously generate or optimize academic schedules.

1. Introduction

University timetable management is an essential academic and administrative activity that directly affects the organization of teaching and learning. A timetable must coordinate multiple elements, including teachers, courses, semesters, academic years, sections, and available time slots. As the number of departments, courses, and teaching staff increases, preparing and maintaining timetables becomes increasingly difficult, particularly when the same faculty member is responsible for teaching courses in more than one department.

In many university environments, timetable preparation is handled independently by individual departments. Although departmental management can simplify local scheduling, it can create coordination challenges when academic resources are shared across departments. A teacher assigned to courses in multiple departments may be scheduled for different classes at the same time if departmental timetables are prepared without centralized coordination. Identifying and resolving such conflicts manually can require repeated communication between departments and may increase the administrative effort involved in timetable preparation.

The problem becomes more significant when timetable information is distributed across separate administrative processes. Different academic authorities may be responsible for managing departments, teachers, courses, and timetable approvals, making it necessary to maintain a structured workflow for creating, reviewing, and approving timetables. A centralized system can provide a common platform through which these activities can be coordinated while preserving appropriate responsibilities for different administrative roles.

Recent advances in artificial intelligence (AI) have also introduced new possibilities for interacting with information systems through natural language. Instead of requiring users to navigate multiple interfaces or manually identify timetable parameters, an AI-assisted interface can interpret natural-language questions and identify relevant information from a user's request. However, integrating AI into an information management system requires careful separation between language interpretation and actual data operations to maintain control over how information is retrieved.

This paper presents an AI-assisted university timetable management system that combines centralized timetable administration with role-based access control, rule-based conflict validation, timetable approval workflows, and natural-language information retrieval. The system is implemented using the MERN stack, with React used for the frontend, Node.js and Express.js for backend services, and MongoDB for data management. The AI-assisted query component uses the Google Gemini API to interpret natural-language timetable queries and extract relevant entities. The backend subsequently uses these structured results to perform

deterministic database queries and return the requested information.

The primary focus of the proposed system is not autonomous timetable generation or schedule optimization. Instead, it aims to provide a centralized environment for managing university timetable information while using AI as an assistance layer for querying that information. By combining conventional rule-based application logic with AI-assisted natural-language interaction, the system provides an architecture for managing shared-faculty scheduling challenges while maintaining explicit control over timetable data and administrative workflows.

2. Problem Statement

University timetable preparation becomes increasingly challenging when academic departments operate independently while sharing teaching resources. In a decentralized scheduling environment, each department may prepare its timetable according to its own courses, sections, and faculty requirements. However, when a teacher teaches courses in multiple departments, the absence of centralized coordination can result in the same teacher being assigned to different classes during the same time slot. Resolving such conflicts through manual communication and repeated timetable revisions increases administrative effort and makes timetable management more difficult to maintain.

Furthermore, timetable information may need to pass through multiple academic authorities before it is finalized, creating a need for a structured process for timetable creation, review, approval, and publication. Another challenge is the way users interact with timetable information. Conventional timetable systems generally require users to navigate predefined forms, filters, or dropdown menus to locate specific information. While these interfaces are useful for structured data entry and management, they may require multiple steps when users simply want to ask questions about teachers, sections, semesters, or departmental schedules.

Therefore, there is a need for a centralized timetable management platform that can coordinate shared academic resources, validate relevant scheduling conflicts during timetable creation, provide controlled administrative workflows, and support more natural interaction with timetable information. The proposed system addresses these requirements by combining centralized management, role-based access control, rule-based conflict validation, timetable approval workflows, and an AI-assisted natural-language query mechanism.

3. Research Objectives

The main objective of this study is to develop a centralized and AI-assisted university timetable management system capable of supporting timetable creation, coordination, validation, approval, and information retrieval. The specific objectives are:

1. **To centralize timetable management** by providing a common platform for managing departments, teachers, courses, semesters, sections, and timetable slots.
2. **To reduce shared-faculty scheduling conflicts** by validating teacher availability during timetable slot creation and preventing conflicting teacher assignments.
3. **To manage batch and section scheduling conflicts** by validating timetable assignments against existing schedules.
4. **To implement role-based access control** so that administrative and academic responsibilities are separated among Super Administrators, Deans, and Chairmen.
5. **To establish a structured timetable approval workflow** in which departmental timetables can be submitted for review and subsequently approved or rejected by the relevant academic authority.
6. **To provide AI-assisted timetable information retrieval** through a natural-language interface that allows users to query timetable information without relying exclusively on predefined navigation and filtering options.
7. **To integrate natural-language interpretation with deterministic database operations** by using the Gemini API to extract query intent and entities while allowing the backend to control actual database operations.
8. **To develop a web-based system using modern full-stack technologies** that provides a maintainable architecture for university timetable administration and information retrieval.

4. Related Work

University course timetabling has been extensively studied as a complex scheduling problem involving multiple hard and soft constraints. These constraints may include instructor availability, student requirements, time-slot assignments, and resource limitations. Chen et al. describe the University Course Timetabling Problem (UCTTP) as a difficult combinatorial optimization problem and review a broad range of methodologies developed to address it [1]. Recent research has also examined increasingly realistic university timetabling environments, including the combination of student sectioning with course time and room assignments in real-world academic settings [2].

Several studies have investigated centralized and distributed approaches to university timetable management. Dimopoulou and Miliotis presented a university course timetabling system based on a centralized database in which different departments could access and update relevant information [3]. Their system also considered common courses involving multiple departments and teacher availability during timetable construction. This work demonstrates the importance of sharing timetable-related information across departmental boundaries when academic resources are not isolated within individual departments.

Inter-department scheduling conflicts have also been investigated using computational and intelligent scheduling approaches. Nourmohammadi-Khiarak et al. proposed a multi-agent method for controlling collisions involving common events between university departments [4].

Their approach considered coordination between separately generated departmental schedules in order to reduce conflicts involving shared academic resources. Such research demonstrates that coordination between departmental scheduling processes can be an important aspect of university timetable management.

In addition to centralized coordination, existing research has investigated automated timetable construction and optimization. These approaches commonly formulate timetabling as a constraint satisfaction, mathematical optimization, or heuristic search problem [1], [2], [4]. Such methods can be effective for constructing schedules under multiple constraints; however, their primary focus is generally the computational generation or optimization of timetable solutions.

Natural-language interaction with databases represents another relevant research direction. Recent surveys have examined the use of Large Language Models (LLMs) for natural-language-to-database-query tasks, including user-intent interpretation, schema understanding, query generation, and database interaction [5]–[7]. These developments demonstrate the potential of language models to provide more accessible interfaces for database-driven applications. At the same time, natural-language database interaction introduces challenges related to ambiguity, schema interpretation, query correctness, and controlled data access [5]–[7].

Role-based access control is also an established approach for separating system permissions according to user responsibilities. Ferraiolo et al. presented an RBAC model and reference implementation for managing authorization within networked systems [8]. The separation of responsibilities provided by RBAC is particularly relevant to applications in which different administrative users must access different categories of data and operations.

The proposed system adopts a different approach by using an AI model primarily as an interpretation layer rather than as an autonomous timetable generator or unrestricted database-query executor. In the implemented architecture, Gemini is used to interpret natural-language timetable questions and extract relevant entities, while the Node.js/Express backend performs controlled database operations through predefined application logic and MongoDB/Mongoose queries. This separation maintains application-level control over timetable data while providing users with a natural-language interaction mechanism.

Therefore, the proposed system builds upon existing research in university timetabling, inter-department scheduling, role-based access control, and AI-assisted database interaction. Its focus is the combination of centralized timetable management, rule-based validation of teacher and batch/section conflicts, role-based administrative control, timetable approval workflows, and AI-assisted natural-language information retrieval within a single web-based platform.

5. Proposed System

The proposed system is a centralized web-based university timetable management platform designed to coordinate timetable-related activities across academic departments. The system combines role-based administrative management, timetable creation, rule-based conflict validation, timetable approval, and AI-assisted natural-language information retrieval within a single application.

5.1 System Roles and Responsibilities

The proposed system defines multiple administrative roles to support the university timetable workflow.

- **Super Administrator:** Manages university-level administrative information. This role is responsible for creating and managing Deans and managing faculties, including assigning Deans to their respective faculties. The Super Administrator can also access approved timetable information through the system dashboard.
- **Dean:** Operates at the faculty level. This role manages Chairmen and departments within the assigned faculty and is responsible for reviewing submitted timetables. Timetables submitted by departments are presented to the Dean for review, after which the Dean can approve or reject them.
- **Chairman:** Responsible for departmental academic and timetable management. This role manages teachers, sections or batches, semesters, courses, and timetable slots within the department. The Chairman can create and manage departmental timetables while the system validates relevant scheduling conflicts during slot assignment.
- **Guest/User:** The system also provides timetable information to users who do not participate in administrative management. Users can navigate timetable information through structured selections and can interact with the AI-assisted query interface to retrieve relevant timetable information.

The role separation follows the general principle of role-based authorization, where permissions are associated with defined responsibilities rather than being granted uniformly to all users [8].

5.2 Centralized Timetable Management

The system centralizes timetable information that would otherwise be managed through separate departmental processes. Academic information such as departments, teachers, courses, semesters, sections, and timetable slots is maintained within a common database.

When a Chairman creates a timetable slot, the backend validates the assignment against existing timetable information. This allows the system to identify relevant conflicts during the timetable creation process rather than relying solely on manual inspection after the timetable

has been completed.

The conflict-validation mechanism particularly addresses shared-faculty scheduling. If a teacher has already been assigned to another class during the selected time period, the system prevents the conflicting assignment and provides an appropriate validation response. The system also validates batch and section scheduling constraints so that conflicting assignments involving the same academic group can be identified during timetable creation.

The current implementation focuses on teacher and batch/section conflicts. Room or classroom conflict validation is not included in the current version.

5.3 Timetable Approval Workflow

Timetable creation and timetable approval are treated as separate stages within the system. After a Chairman prepares a departmental timetable, it can be submitted for review by the relevant Dean.

The Dean interface provides access to pending timetables organized according to relevant academic information, such as faculty, department, semester, and academic year. The Dean can inspect the timetable details and either approve or reject the submission.

This workflow introduces an additional administrative verification stage before a timetable becomes part of the approved timetable information available through the system.

5.4 AI-Assisted Query Architecture

The system incorporates an AI-assisted query mechanism to provide a natural-language interface for timetable information. Rather than allowing the language model to directly control database operations, the architecture separates natural-language interpretation from data retrieval.

When a user submits a natural-language question, the query is sent to the AI processing component. The Google Gemini API is used to interpret the query and extract structured information relevant to timetable retrieval. Depending on the request, extracted information may include the department, academic year, semester, section, or teacher name.

The resulting structured representation is then processed by the backend. The backend determines the appropriate database operation and executes the corresponding query against MongoDB using application-level logic and Mongoose. The retrieved information is subsequently returned to the user through the application interface.

This design means that Gemini serves as an interpretation component rather than a direct database controller. Database access remains within the backend application logic.

6. System Architecture and Workflow

The proposed system follows a layered client-server architecture in which the presentation layer, application layer, data layer, and AI-assisted query component perform separate responsibilities.

Figure 1. Three-Level System Architecture

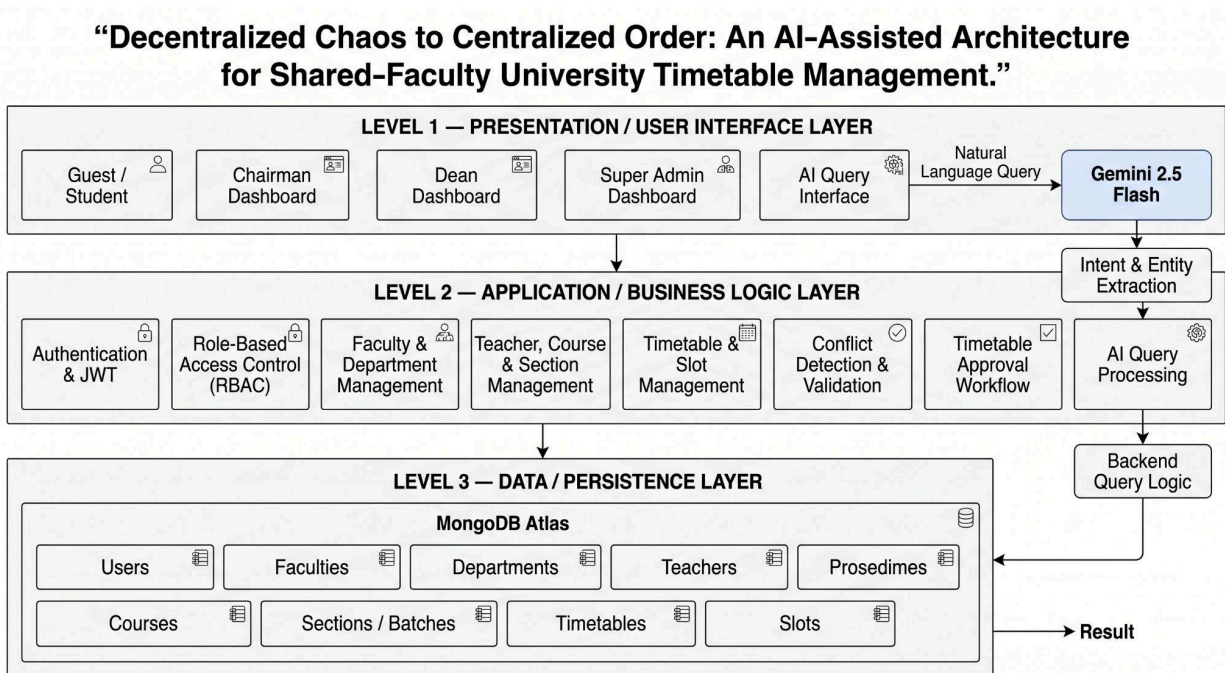


Figure 1. Three-Level System Architecture of the proposed timetable management system showing the Presentation/User Interface, Application/Business Logic, and Data/Persistence layers.

The presentation layer contains the user-facing interfaces, including Guest/User timetable access, Chairman Dashboard, Dean Dashboard, Super Administrator Dashboard, and the AI Query Interface.

The application layer contains authentication, authorization, faculty and department management, teacher/course/section management, timetable and slot management, conflict validation, timetable approval, and AI query processing.

The data layer contains MongoDB Atlas and the primary application data, including users, faculties, departments, teachers, courses, sections/batches, timetables, and timetable slots.

The AI query component interacts with the application layer rather than directly accessing the

database. Natural-language input is interpreted by Gemini, after which the backend processes the extracted information and performs the corresponding database operation.

6.1 Complete Administrative Workflow

The creation and approval of university timetables follows a hierarchical administrative workflow.

Figure 2. University Timetable Management and Approval Workflow

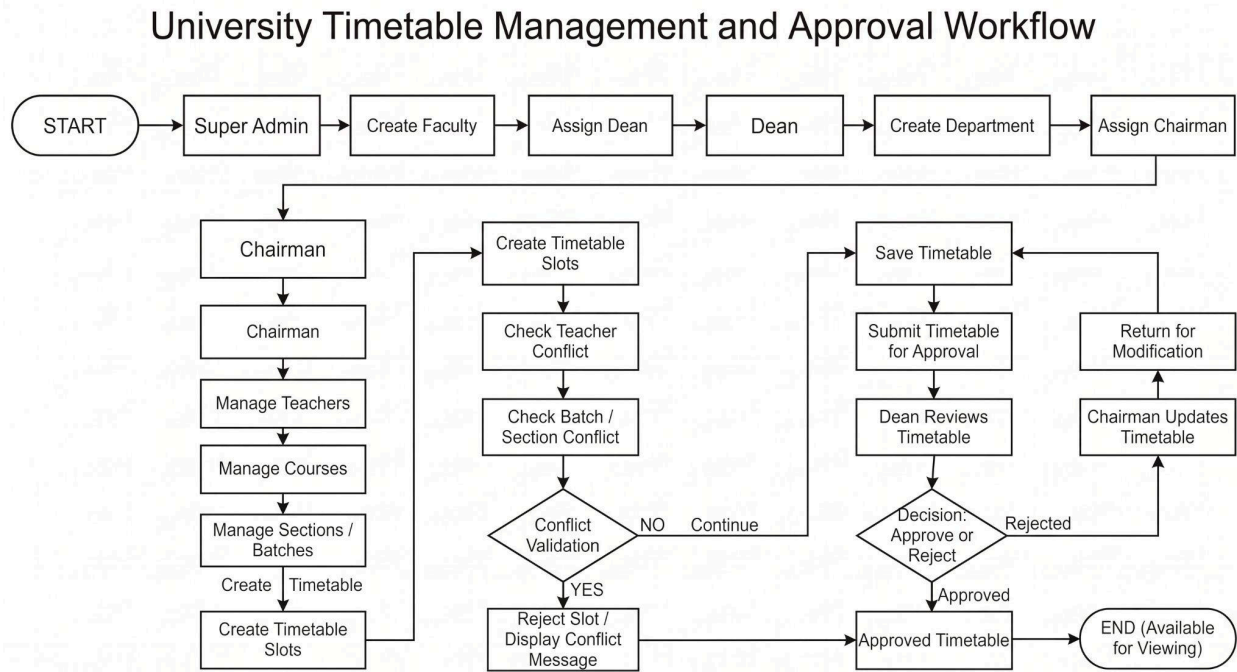


Figure 2. Complete university timetable management and approval workflow from faculty and department configuration to timetable creation, conflict validation, and Dean approval.

The workflow begins with the Super Administrator creating faculties and assigning Deans. The Dean then manages departments and assigns Chairmen. After assignment, the Chairman manages departmental teachers, courses, sections or batches, and timetable information.

During timetable creation, each slot is passed through the conflict-validation mechanism. If a scheduling conflict is detected, the slot is rejected and the user is required to modify the assignment. If no relevant conflict is detected, the slot is saved and the timetable can subsequently be submitted for approval.

The Dean reviews the submitted timetable. An approved timetable becomes available through the approved timetable interface, while a rejected timetable is returned to the Chairman for

modification.

6.2 Timetable Slot Conflict Detection and Validation

During slot allocation by a department Chairman, rule-based validation is performed deterministically.

Figure 3. Timetable Slot Conflict Detection and Validation

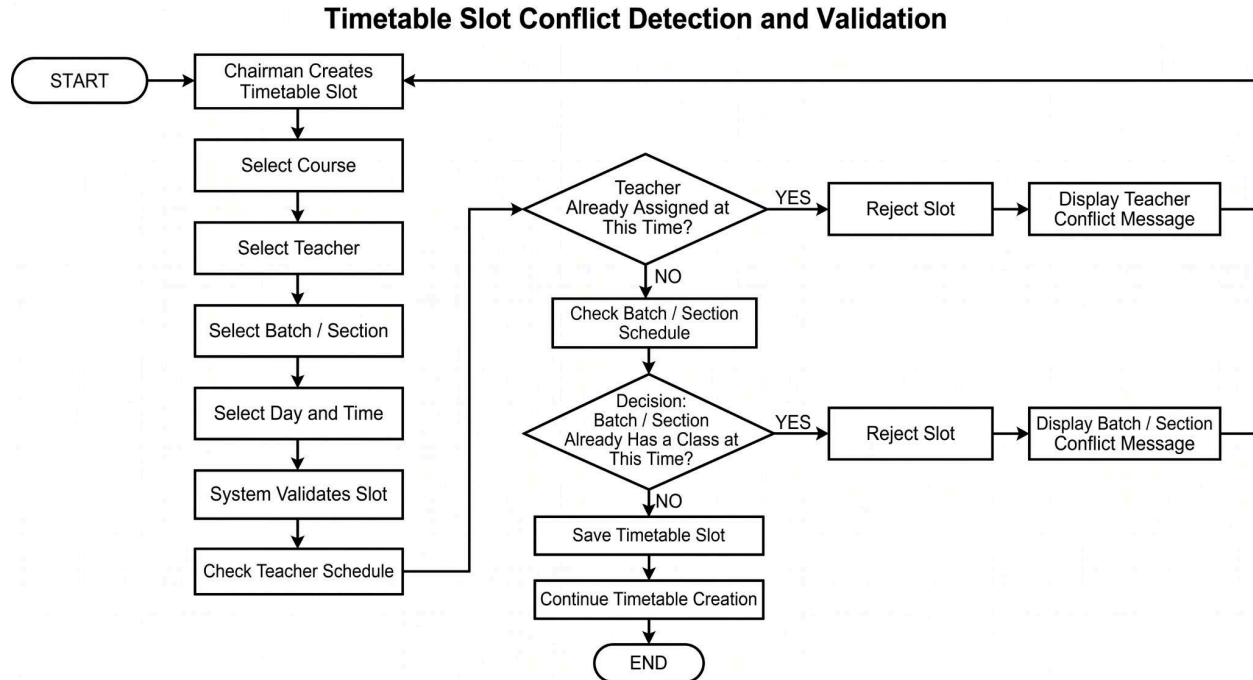


Figure 3. Timetable slot conflict detection and validation flowchart for teacher and batch/section scheduling constraints.

The validation process begins when a Chairman creates a timetable slot by selecting a course, teacher, batch or section, and time information. The system first checks whether the selected teacher already has an assignment during the specified time period.

If a teacher conflict is detected, the slot is rejected and an error response is displayed. If no teacher conflict exists, the system continues to check the relevant batch or section schedule. If the batch or section is already occupied during the selected time, the slot is rejected. Otherwise, the slot is saved.

The current validation mechanism therefore focuses on two implemented constraints: teacher scheduling conflicts and batch/section scheduling conflicts. Classroom or room availability is not included in the current implementation.

6.3 AI-Assisted Query Pipeline

The AI-assisted query subsystem separates natural-language interpretation from database operations.

Figure 4. AI-Assisted Natural-Language Query Processing Architecture

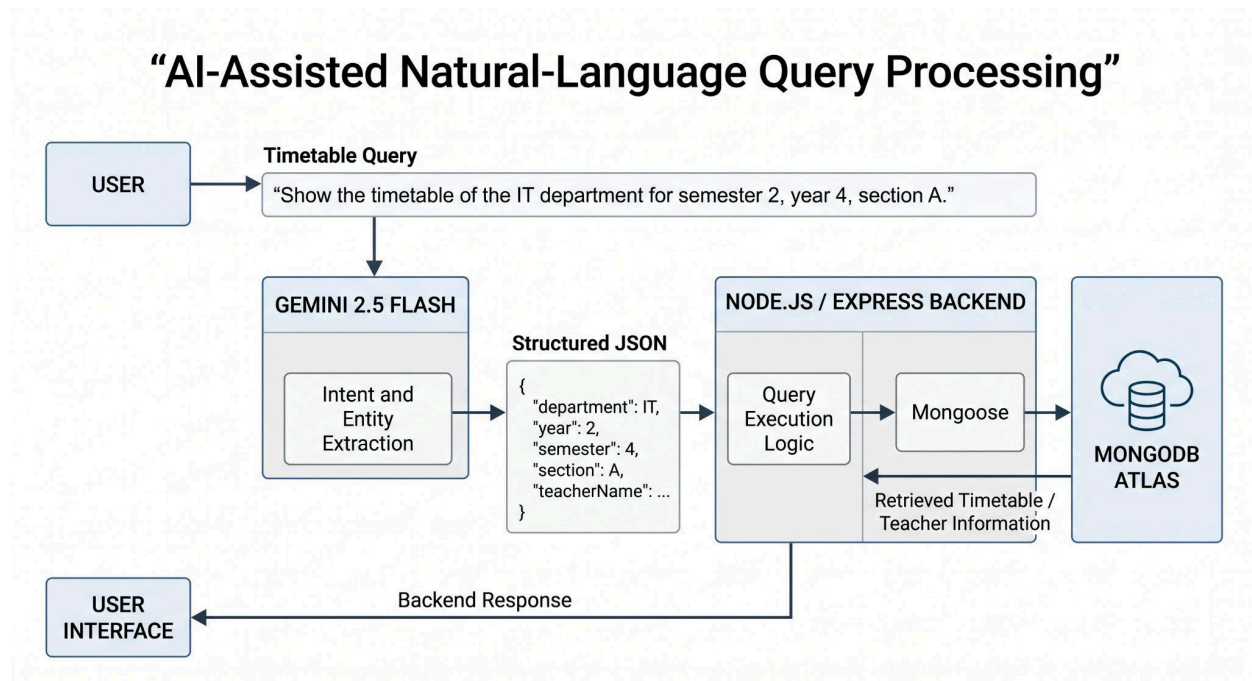


Figure 4. AI-assisted natural-language query processing architecture showing the separation between Gemini-based interpretation and backend database operations.

A user can submit a natural-language query such as:

"Show IT department year 4 semester 2 section A timetable."

The Gemini 2.5 Flash model interprets the query and extracts relevant entities. An example structured representation may contain:

```
{  
  
  "department": "IT",  
  
  "year": 4,  
  
  "semester": 2,
```

```
"section": "A",  
"teacherName": null  
}
```

The structured information is then passed to the Node.js/Express backend. The backend determines the appropriate query execution logic and uses Mongoose to retrieve the relevant information from MongoDB Atlas.

The retrieved information is returned through the backend response and displayed through the application interface. Gemini therefore functions as an interpretation layer rather than directly modifying or querying the database.

7. Results and System Validation

The developed system was evaluated through functional validation of its major implemented components. The evaluation focused on administrative role management, departmental timetable management, teacher and batch/section conflict validation, timetable approval, and AI-assisted timetable information retrieval.

The results presented in this section are based on representative interfaces and functional behavior of the implemented system. The evaluation does not claim large-scale statistical performance measurements; instead, it demonstrates that the major intended workflows and features are implemented and operational within the developed application.

7.1 Administrative Management

The Super Administrator provides university-level management of faculties and Deans. The dashboard provides access to the administrative operations required to establish the faculty-level structure of the system.

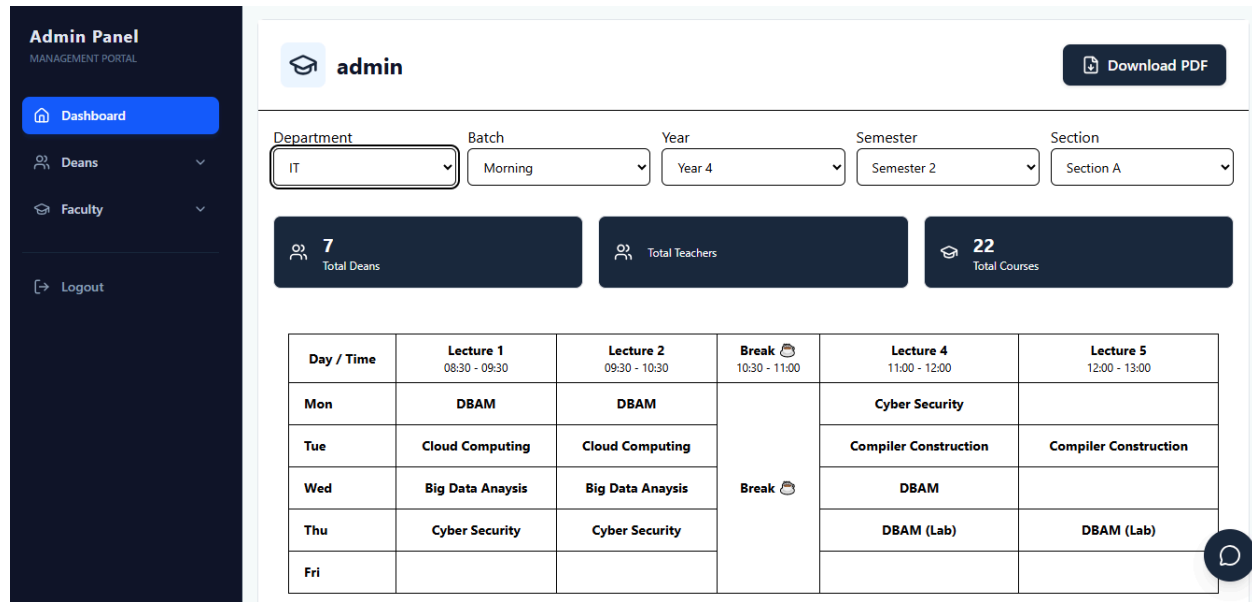


Figure 5. Super Administrator dashboard for managing faculties and Deans.

The Dean interface provides faculty-level management functions, including department and Chairman management. This creates the administrative relationship required for subsequent timetable submission and approval.

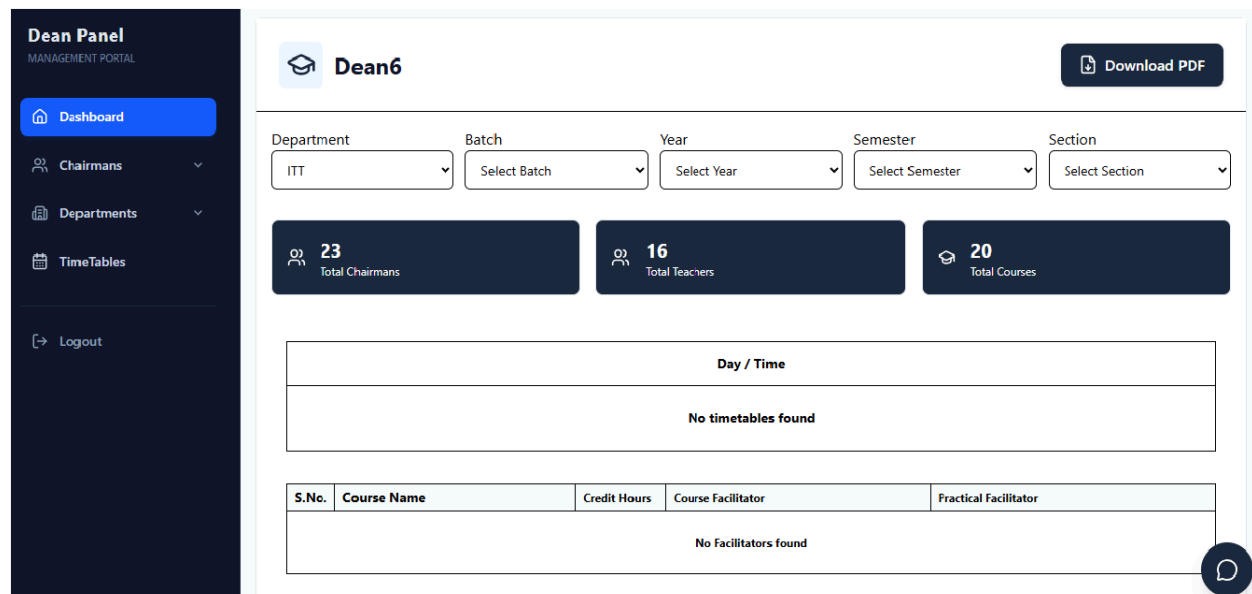


Figure 6. Dean interface for managing departments and assigning Chairmen.

These interfaces demonstrate the separation of responsibilities between university-level and faculty-level administrative operations.

7.2 Departmental Timetable Management

The Chairman is responsible for managing departmental academic information and creating timetable schedules. The Chairman dashboard provides access to teacher, course, section/batch, semester, timetable, and slot management operations.

The screenshot displays the 'Chairman Panel' Management Portal. On the left is a dark sidebar with navigation links: Dashboard (highlighted), Teachers, Sections & Batches, TimeTable, Semester, Courses, View Teacher Slots, and a Logout button. The main content area is titled 'Guest Chairman' for the 'Guest Department'. It features a 'Download PDF' button and four filter dropdowns: Batch (Select Batch), Year (Select Year), Semester (Select Semester), and Section (Select Section). Below these are three summary cards: '10 Total Slots', '6 Total Teachers', and '5 Total Courses'. A table section titled 'Day / Time' shows 'No timetables found'. Another table with headers 'S.No.', 'Course Name', 'Credit Hours', 'Course Facilitator', and 'Practical Facilitator' shows 'No Facilitators found'. A notification bell icon is in the bottom right corner.

Figure 7. Chairman dashboard for departmental academic and timetable management.

After the required academic information has been configured, the Chairman can create timetable slots by selecting the relevant course, teacher, section or batch, and time information.

Chairman Panel
MANAGEMENT PORTAL

- Dashboard
- Teachers
- Sections & Batches
- TimeTable**
 - Create TimeTable
 - Manage TimeTable
 - Manage Slots
- Semester
- Courses
- View Teacher Slots

Create Timetable Slot Guest Department

Time Tables: morning, Sem No 1 Year No1

Courses: Guest Course | Guest Teacher

Type: Theory

Teacher: teacher6, ITT

Day: Monday

Start Time: 08:30 AM

End Time: 09:30 AM

+ Create Timetable Slot

Figure 8. Timetable slot creation interface used by the Chairman.

The slot-creation interface provides the entry point for the rule-based validation mechanism described in Section 6.2.

7.3 Conflict Detection and Validation

Conflict validation is one of the central functional features of the developed system. The implementation checks whether a selected teacher is already assigned to another class during the selected time period. When such a conflict is identified, the system prevents the conflicting slot from being created.

Chairman Panel
MANAGEMENT PORTAL

- Dashboard
- Teachers
- Sections & Batches
- TimeTable**
 - Create TimeTable
 - Manage TimeTable
 - Manage Slots
- Semester
- Courses
- View Teacher Slots

Create Timetable Slot Guest Department

Teacher already has a class on this time

Time Tables: morning, Sem No 2 Year No2

Courses: Guest Course | Guest Teacher

Type: Theory

Teacher: Prof. Dr.M.Suleman Memon, IT

Day: Monday

Start Time: 08:30 AM

End Time: 09:30 AM

+ Create Timetable Slot

Figure 9. Teacher scheduling conflict detected during timetable slot creation.

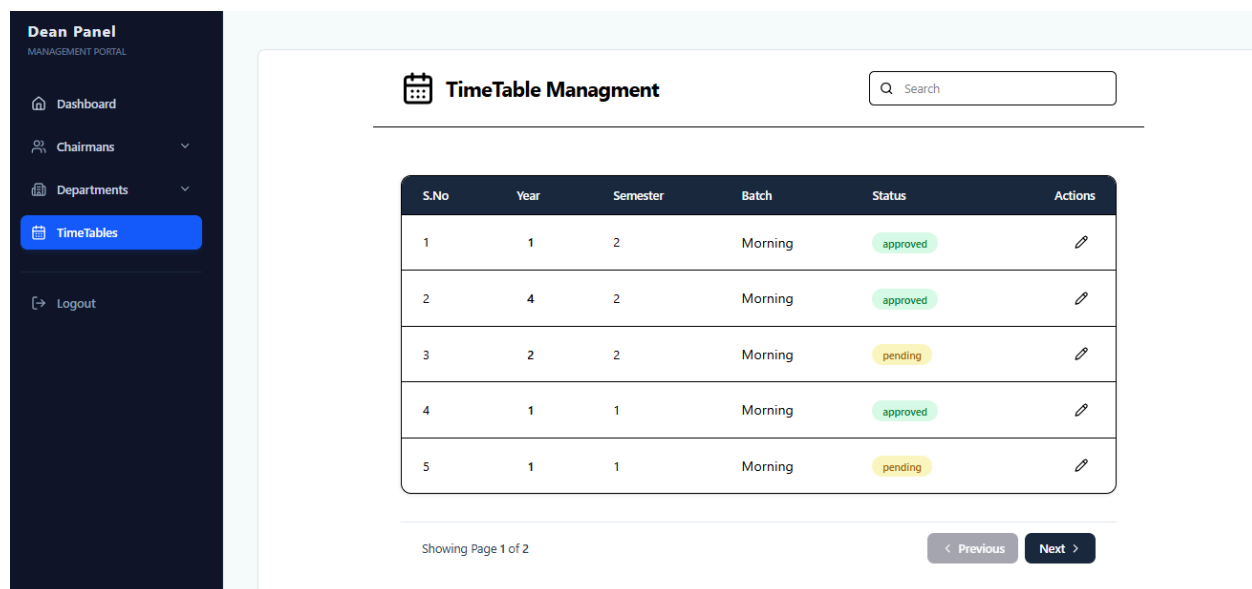
A corresponding validation mechanism is also applied to batch and section schedules. When the selected academic group already has a timetable assignment during the selected time period, the system prevents the conflicting assignment.

These results demonstrate that the implemented system performs validation during timetable creation rather than relying solely on manual inspection after timetable preparation.

The current implementation does not validate room or classroom conflicts. Therefore, such constraints are outside the scope of the reported results.

7.4 Timetable Approval Workflow

After timetable creation, the Chairman can submit the timetable for review. Submitted timetables become available to the relevant Dean through the pending timetable interface.



The screenshot displays the 'Dean Panel' interface, which is a management portal. On the left is a dark sidebar with navigation links: Dashboard, Chairmans, Departments, TimeTables (highlighted in blue), and Logout. The main content area is titled 'TimeTable Managment' and features a search bar. Below the header is a table with the following data:

S.No	Year	Semester	Batch	Status	Actions
1	1	2	Morning	approved	
2	4	2	Morning	approved	
3	2	2	Morning	pending	
4	1	1	Morning	approved	
5	1	1	Morning	pending	

At the bottom of the table, it indicates 'Showing Page 1 of 2' and includes 'Previous' and 'Next' navigation buttons.

Figure 10. Dean interface displaying submitted timetables pending approval.

The Dean can inspect the submitted timetable and make an approval or rejection decision.

Dean Panel

MANAGEMENT PORTAL

Dashboard

Chairmans

Departments

TimeTables

Logout

←

Timetable Details

Department: IT

Semester: 2

Year: 4

Batch: morning

Section: A

Faculty: Engineering

approved

Day / Time	Lecture 1 08:30 - 09:30	Lecture 2 09:30 - 10:30	Break 10:30 - 11:00	Lecture 4 11:00 - 12:00	Lecture 5 12:00 - 13:00
Mon	DBAM	DBAM	Break	Cyber Security	
Tue	Cloud Computing	Cloud Computing		Compiler Construction	Compiler Construction
Wed	Big Data Anaysis	Big Data Anaysis		DBAM	
Thu	Cyber Security	Cyber Security		DBAM (Lab)	DBAM (Lab)
Fri					

Figure 11. Timetable review interface providing approval and rejection actions.

When a timetable is approved, it becomes available through the approved timetable information. If it is rejected, the Chairman can modify the timetable and resubmit it.

This workflow provides a clear separation between timetable creation and timetable approval and ensures that departmental timetable information passes through an additional administrative review stage.

7.5 AI-Assisted Natural-Language Query

The developed system provides an AI-assisted interface through which users can query timetable information using natural language. The user does not need to manually navigate through every timetable parameter when making a supported query.


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Login

Time Table Management System

Select your Department, Batch, Year, Semester to continue

Department

Batch

Year

Semester

Data Science

Select Batch

Select Year

Select Semester

Day / Time

No timetables found

S.No.	Course Name	Credit Hours	Course Facilitator	Practical
No Facilitators found				

UniTime Assistant

Hello! Ask me to fetch timetable of any department by giving semester , year, batch, and department name .

Ask something...

Figure 12. AI-assisted natural-language query interface for timetable information retrieval.

For example, a user can submit a query specifying a department, year, semester, section, or teacher, batch. Gemini interprets the query and extracts the relevant entities. The backend subsequently uses those entities to perform a controlled database query.


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Login

Time Table Management System

Select your Department, Batch, Year, Semester to continue

Day / Time	Lecture 1 08:30 - 09:30	Lecture 2 09:30 - 10:30	Break 10:30 - 11:00	Lecture 4 11:00 - 12:00
Mon	DBAM	DBAM	Break	Cyber Security
Tue	Cloud Computing	Cloud Computing		Compiler Constructi
Wed	Big Data Anaysis	Big Data Anaysis		DBAM
Thu	Cyber Security	Cyber Security		
Fri				

S.No.	Course Name	Credit Hours	Course Facilitator
01	DBAM	3 + 1	Prof. Dr.M.Suleman Memon
02	Cyber Security	3 + 0	Prof. Dr.Mukhtair Ahmed Memon

UniTime Assistant

Show me the timetable of IT department, 4th year 2 semester morning section A

IT timetable for semester 2 year 4 batch morning fetched successfully

Ask something...

Figure 13. Timetable information retrieved through the AI-assisted query mechanism.

The AI component therefore improves the interaction method for retrieving timetable

information while leaving actual database operations under backend application control.

7.6 Timetable Viewing

The final timetable can be displayed to users through the timetable viewing interface. Users can access timetable information according to the available structured selections and approved timetable data.

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[Login](#)

Time Table Management System

Select your Department, Batch, Year, Semester to continue

[Download PDF](#)

Department: IT Batch: Morning Year: Year 4 Semester: Semester 2 Section: Section A

Day / Time	Lecture 1 08:30 - 09:30	Lecture 2 09:30 - 10:30	Break 10:30 - 11:00	Lecture 4 11:00 - 12:00	Lecture 5 12:00 - 13:00
Mon	DBAM	DBAM	Break	Cyber Security	
Tue	Cloud Computing	Cloud Computing		Compiler Construction	Compiler Construction
Wed	Big Data Analysis	Big Data Analysis		DBAM	
Thu	Cyber Security	Cyber Security		DBAM (Lab)	DBAM (Lab)
Fri					

S.No.	Course Name	Credit Hours	Course Facilitator	Practical Facilitator
01	DBAM	3 + 1	Prof. Dr.M.Suleman Memon	Mr Abdul Hafeez

Figure 14. Approved timetable displayed to the user through the timetable viewing interface.

The displayed timetable represents the final output of the implemented administrative workflow, from academic information management and slot creation to conflict validation and approval.

7.7 Summary of Functional Validation

The functional validation demonstrates the major capabilities implemented in the system.

System Feature	Validation Result
Faculty and Dean management	Implemented and demonstrated
Department and Chairman management	Implemented and demonstrated

Teacher management	Implemented
Course management	Implemented
Section/Batch management	Implemented
Timetable and slot creation	Implemented and demonstrated
Teacher conflict validation	Implemented and demonstrated
Batch/section conflict validation	Implemented and demonstrated
Room/classroom conflict validation	Not implemented
Timetable submission	Implemented
Dean timetable review	Implemented and demonstrated
Timetable approval/rejection	Implemented and demonstrated
AI-assisted natural-language querying	Implemented and demonstrated
Autonomous timetable generation	Not implemented
Automated timetable optimization	Not implemented

The results indicate that the developed application provides the core centralized management, validation, administrative approval, and AI-assisted information retrieval capabilities defined in the research objectives. The current evaluation is functional rather than a large-scale

quantitative assessment of usability, scalability, or optimization quality.

8. Limitations and Future Work

Although the proposed system provides centralized timetable management, conflict validation, role-based administration, and AI-assisted information retrieval, several limitations remain in the current implementation.

First, the conflict detection mechanism primarily addresses teacher and batch/section scheduling conflicts. The current version does not include room or classroom conflict validation. Future versions can extend the validation layer to include classroom availability, room capacity, equipment requirements, and other resource constraints.

Second, the system does not automatically generate or optimize complete timetables. Timetable creation remains a controlled administrative process in which authorized users assign courses, teachers, and time slots. Future research could investigate automated timetable generation and optimization techniques that consider multiple constraints simultaneously while retaining administrative control over the final schedule.

Third, the AI component is focused on natural-language query interpretation and information retrieval rather than autonomous timetable generation. Gemini is used to interpret user queries and extract relevant entities, while the backend performs the actual database operations. Future work could expand the conversational interface to support more complex multi-step queries, contextual follow-up questions, and richer timetable analysis while maintaining backend validation and access-control restrictions.

The current system also depends on the availability of the external Gemini API for AI-assisted queries. Temporary service failures or rate-limit responses may affect AI query processing. Future implementations could introduce additional fallback mechanisms or alternative models to improve availability and resilience.

Finally, the current evaluation primarily focuses on functional validation of the implemented features. A larger-scale evaluation involving multiple departments, teachers, and students could provide quantitative evidence regarding usability, response time, conflict-detection effectiveness, and system scalability.

Future development can therefore focus on expanding constraint validation, introducing optional timetable generation and optimization, improving conversational query capabilities, strengthening service resilience, and conducting broader empirical evaluation. These extensions would build upon the existing centralized architecture while preserving the system's rule-validated and role-controlled design.

9. Conclusion

This study presented an AI-assisted university timetable management system designed to address coordination challenges associated with decentralized departmental scheduling, particularly when teaching staff are shared across multiple departments. The proposed architecture centralizes timetable information and administrative workflows, enabling authorized users to create, manage, validate, and approve timetables within a unified platform.

A key feature of the system is its rule-based conflict validation, which checks teacher and batch/section scheduling constraints during timetable creation. This approach helps identify relevant scheduling conflicts before a timetable is submitted for approval. The role-based access control model further separates administrative responsibilities among Super Administrators, Deans, and Chairmen, while the approval workflow provides an additional level of administrative review.

The system also incorporates an AI-assisted natural-language query mechanism using Google Gemini. Rather than allowing the AI model to directly manipulate timetable data, the system uses the model to interpret natural-language queries and extract relevant entities. The backend then performs controlled database queries using the extracted information. This architecture combines the flexibility of natural-language interaction with deterministic backend processing and application-level control.

Overall, the developed system demonstrates how centralized data management, rule-based validation, role-based administration, and selective AI assistance can be combined within a MERN-stack application for university timetable management. The implemented system helps prevent teacher and batch/section scheduling conflicts during timetable creation while providing a structured administrative workflow for timetable review and approval.

The current implementation establishes a foundation for future extensions involving broader resource constraints, automated timetable generation, richer conversational queries, and larger-scale system evaluation. These extensions can further enhance the platform while maintaining administrative control over scheduling decisions.

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